

Customer Churn Analysis

Project Report

Tools: Python | SQL | Power BI

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1. About the Project

Customer churn means a customer stops buying from a business. This project studies a customer dataset to find out how many customers left, which customers are more likely to leave, and why.

- Find the overall churn rate.
- Find which customer groups churn the most.
- Find the behaviours and reasons linked to churn.
- Suggest simple actions to keep more customers.

Python was used to clean and explore the data, SQL to summarise sales and customer scores, and Power BI to build the dashboard.

2. About the Dataset

The dataset has **50,000 customers** (one row per customer) and **17 columns**. It has customer details, buying behaviour, satisfaction scores and whether the customer left.

Column	Type	Meaning
CustomerID	Text	Unique customer code
JoinDate, LastPurchaseDate	Date	When the customer joined and last bought
Age, Gender	Number, Text	Age (21 to 67); Female, Male or Other
Region	Text	North, South, East, West or Central
CustomerSegment	Text	Startup, SMB, Mid-Market or Enterprise
ProductCategory	Text	Subscription, Software, Consulting, Services or Hardware
TotalPurchases, TotalSpend	Number	Number of purchases and money spent
AvgOrderValue	Number	Average value of one order
DaysSinceLastPurchase	Number	Days since the customer last bought
NumSupportTickets	Number	Support tickets raised (0 to 19)
SatisfactionScore	Number	Customer rating from 1 to 5
NPSScore	Number	Recommendation score from -100 to +100
Churn	Number	Target column: 1 = left, 0 = stayed
ChurnReason	Text	Why the customer left (only for churned customers)

3. Data Cleaning

The data was cleaned in Python (pandas) before analysis.

Step	What was done
Duplicates	300 duplicate rows were removed (50,300 rows became 50,000).
Missing Age	Filled with the median age.
Missing Region	Filled with the most common region.
Missing AvgOrderValue	Filled with the median of the same segment and purchase count.
Missing Satisfaction and NPS	Filled with the median.
ChurnReason	Customers who stayed were labelled "Not_Churned".
Data types	Dates converted to date type; Age converted to whole numbers.
Age_Group	New column with three equal-size groups: Young (21-37), Adult (38-51), Senior (52-67).

4. Key Numbers

50,000 Customers	15,324 Churned	34,676 Retained	30.65% Churn rate
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Measure	Value
Total customer spend	₹1,248.7 million
Average spend per customer	₹24,974
Average order value	₹1,576
Average satisfaction score (out of 5)	3.01
Average NPS score	-0.03

Subscription (24.8%) and Software (22.0%) bring in the most spend. Hardware is the smallest (12.9%).

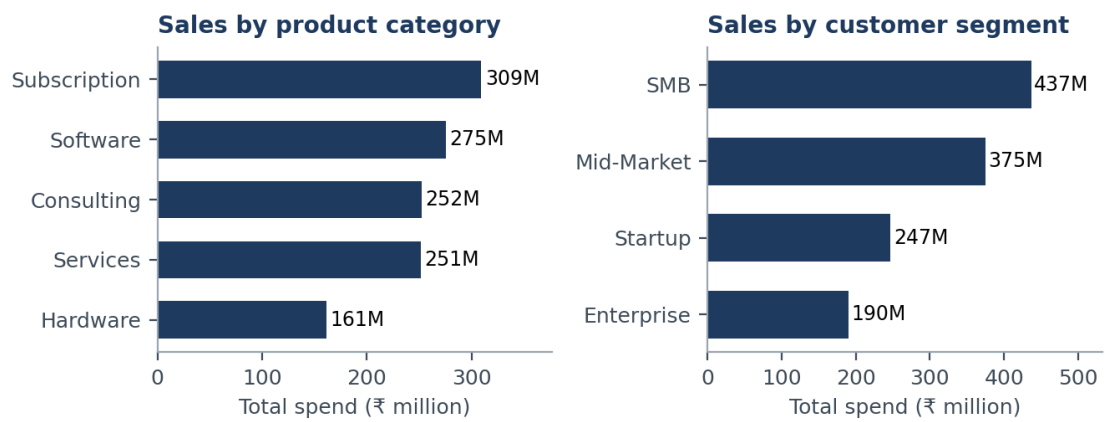


Figure 1. Total customer spend by product category and customer segment.

5. What the Data Shows

5.1 Churn by customer segment

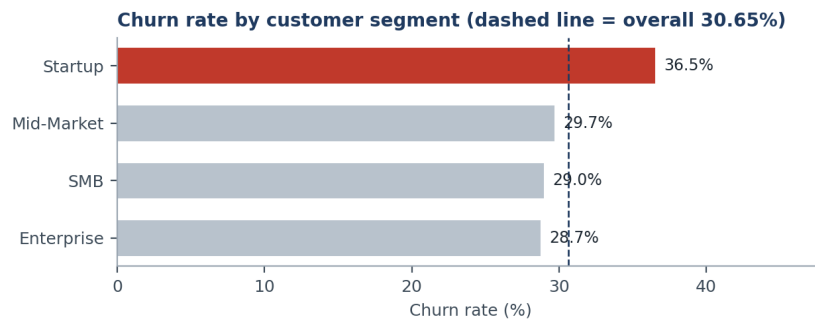


Figure 2. Churn rate by customer segment.

Startup customers churn the most at **36.5%**. The other three segments are between 28.7% and 29.7%.

SMB has the most churned customers in number only because it is the largest segment. Its churn rate is one of the lowest.

5.2 Age, gender, region and product

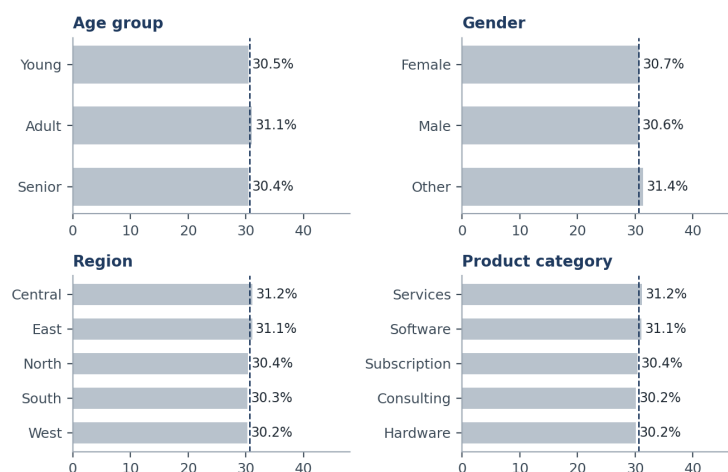


Figure 3. Churn rate is almost the same in every group.

Churn is between 30.2% and 31.4% in every age group, gender, region and product category. These do not separate customers who stay from those who leave.

5.3 Satisfaction and NPS

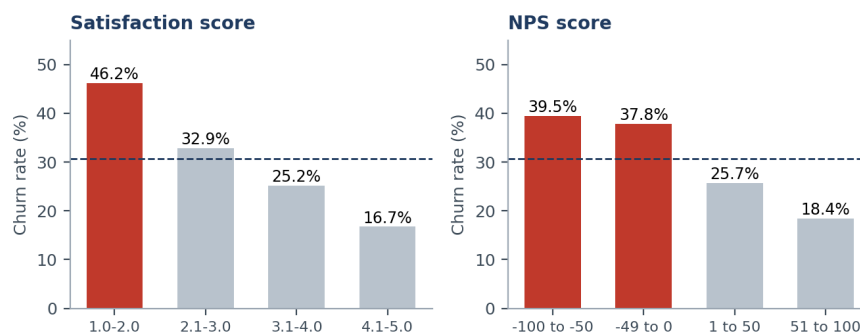


Figure 4. Churn rate by satisfaction score and by NPS score.

Unhappy customers leave more. Churn is **46.2%** for satisfaction 2.0 or below and **16.7%** for satisfaction above 4.0. NPS shows the same pattern.

5.4 Days since last purchase and support tickets

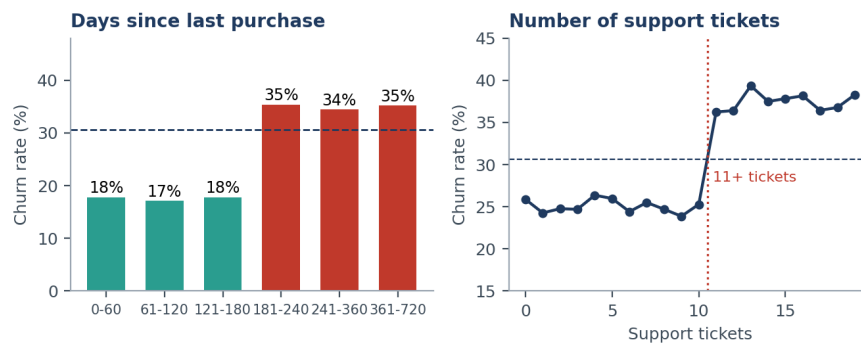


Figure 5. Churn rate by days since last purchase and by support tickets.

- Customers who bought in the last 180 days churn at 17.6%. After 180 days without a purchase, the rate is 35.1%.
- Customers with 10 tickets or fewer churn at 25.1%. With 11 or more tickets, the rate is 37.4%.

5.5 Reasons for churn

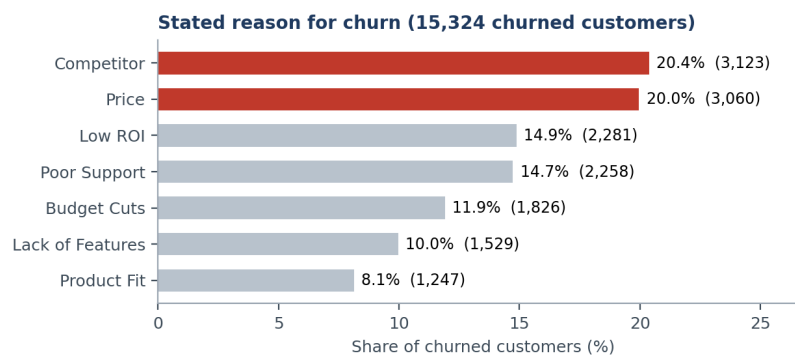


Figure 6. Reason given by churned customers.

Competitor (20.4%) and price (20.0%) are the top two reasons, followed by low ROI (14.9%) and poor support (14.7%).

6. Recommendations

- **Watch inactive customers.** Contact customers before they reach 180 days without a purchase.
- **Follow up on low satisfaction.** Reach out quickly to customers who rate 2.0 or below.
- **Fix support problems.** Escalate customers who raise many tickets (around 8 or more) to a senior person.
- **Plan for Startups.** They churn at 36.5%. Offer better onboarding and flexible pricing, and ask why they leave.
- **Review pricing and competitors.** Together they are about 40% of churn reasons.
- **Track churn rate, not just counts,** by segment in SQL and Power BI.

7. Conclusion

Out of 50,000 customers, 15,324 churned, which is a churn rate of 30.65%. Who the customer is (age, gender, region, product) makes little difference. What the customer does and feels matters more: low satisfaction, a long gap since the last purchase and many support tickets go with higher churn. Startups are the segment with the highest churn (36.5%).

Summary: To reduce churn, focus on unhappy and inactive customers, improve support, and give extra attention to Startup customers.